Lab 7: Fuzzing for Input Validation Bugs - Screenshots

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Step 2: Run the buggy application (python password_policy.py)

```
$ python password_policy.py
==== Running first_character_type ====
Enter a password: First Character Type: letter

==== Running split_credentials ====
Enter user:password (e.g. alice:Secret1): User and Password: ('alice', 'Secret1')

==== Running digit_ratio ====
Enter a password to measure: Digit Ratio: 0.14285714285714285
```

Step 4: Hypothesis tests against the buggy code (failing)

```
$ pytest test_password_policy.py
============================= test session starts ==============================
platform darwin -- Python 3.15.1, pytest-9.2.0, pluggy-1.7.0
rootdir: lab7
plugins: hypothesis-6.170.15
collected 3 items
test_password_policy.py FFF                                              [100%]
=================================== FAILURES ===================================
_____________________ test_first_character_type_no_crash ______________________
  + Exception Group Traceback (most recent call last):
  |   File "lab7/test_password_policy.py", line 5, in test_first_character_type_no_crash
  |     def test_first_character_type_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "lab7/test_password_policy.py", line 7, in test_first_character_type_no_crash
    |     first_character_type(s)
    |     ~~~~~~~~~~~~~~~~~~~~^^^
    |   File "lab7/password_policy.py", line 2, in first_character_type
    |     ch = pw[0]
    |          ~~^^^
    | IndexError: string index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "lab7/test_password_policy.py", line 9, in test_first_character_type_no_crash
    |     assert False, f"first_character_type crashed with: {e!r}"
    | AssertionError: first_character_type crashed with: IndexError('string index out of range')
    | assert False
    | Failing test case: test_first_character_type_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         lab7/test_password_policy.py:8
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "lab7/test_password_policy.py", line 7, in test_first_character_type_no_crash
    |     first_character_type(s)
    |     ~~~~~~~~~~~~~~~~~~~~^^^
    |   File "lab7/password_policy.py", line 2, in first_character_type
    |     ch = pw[0]
    |          ~~^^^
    | TypeError: 'NoneType' object is not subscriptable
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "lab7/test_password_policy.py", line 9, in test_first_character_type_no_crash
    |     assert False, f"first_character_type crashed with: {e!r}"
    | AssertionError: first_character_type crashed with: TypeError("'NoneType' object is not subscrip
table")
    | assert False
    | Failing test case: test_first_character_type_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         lab7/test_password_policy.py:8
    +------------------------------------
```

Step 5: Run the fixed application

```
$ python password_policy.py
==== Running first_character_type ====
Enter a password: First Character Type: letter

==== Running split_credentials ====
Enter user:password (e.g. alice:Secret1): User and Password: ('alice', 'Secret1')

==== Running digit_ratio ====
Enter a password to measure: Digit Ratio: 0.14285714285714285

$ python password_policy.py   # edge-case input
==== Running first_character_type ====
Enter a password: First Character Type: empty

==== Running split_credentials ====
Enter user:password (e.g. alice:Secret1): User and Password: (None, None)

==== Running digit_ratio ====
Enter a password to measure: Digit Ratio: 0.0
```

Step 6: Re-run tests against the fixed code (passing)

```
$ pytest test_password_policy.py
============================= test session starts ==============================
platform darwin -- Python 3.15.1, pytest-9.2.0, pluggy-1.7.0
rootdir: lab7
plugins: hypothesis-6.170.15
collected 3 items
test_password_policy.py ...                                              [100%]
============================== 3 passed in 0.53s ===============================
```